MASTER THESIS

Data-Driven Clustering of Intracranial Pressure Time-Series and CT Characterization

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PROJECT DETAILS

- ***Role*** → AI and Biomedical Data Science Researcher
- ***Supervision & Setting*** → Intensive Care Laboratory, UZ Leuven (Prof. Dr. Geert Meyfroidt, Fadime Tokmak)
- ***Technical Stack*** →
  - Core & Analytics: Python (NumPy, Pandas, SciPy, Scikit-Learn)
  - Time-Series ML: tslearn, PyEMD
  - Data Visualization: Matplotlib, Seaborn
- ***Core Domains*** →
  - Time-series data engineering & cohort cleaning (physiological plausibility, gap handling)
  - Non-stationary signal processing (EMD decomposition)
  - Artifact removal = artifact detection + artifact imputation (MAD filtering, PCHIP interpolation)
  - High-dimensional feature extraction & selection (Mutual Information, Pearson correlation, PCA)
  - Unsupervised feature-based clustering (K-means, hierarchical, DBSCAN, GMM, spectral)
  - Unsupervised waveform-based clustering (Dynamic Time Warping (DTW), Self-Organizing Maps (SOM))
  - Statistical association modeling (Cramér's V analysis)
  - Clinical data interpretation & matrix seriation heatmaps

CLINICAL CONTEXT & PROBLEM STATEMENT

In neuro-intensive care, traumatic brain injury (TBI) remains a leading cause of hospitalizations, long-term disability, and mortality worldwide. Beyond the primary trauma, secondary brain injuries (such as cerebral edema and expanding hemorrhages) develop over the subsequent hours and days, driving adverse neurological outcomes.

While continuous intracranial pressure (ICP) monitoring is the clinical standard of care for severe TBI, standard medical practice relies on reactive management instead of proactive:

- **Limitations of Static Thresholds:** Clinical guidelines historically trigger interventions when ICP exceeds a fixed cutoff or focus on broad arithmetic averages, disregarding signal morphology, temporal evolution, and the accumulated dose of secondary injury.
- **The "Black-Box" Predictive Gap:** Supervised machine learning algorithms can predict patient deterioration with high statistical accuracy, but they operate as black boxes without explaining why patient trajectories diverge or how distinct pressure behaviors map to underlying physical trauma.
- **Disconnection Between Neuroimaging and Physiological Streams:** Computed tomography (CT) scans capture static anatomical damage upon admission (evaluated via the Marshall CT classification), but there isn't a lot of knowledge about how specific structural injury classes translate into continuous dynamic ICP trajectories over time.

The objective

🎯 ***Main goal*** → *explore feasibility / clinical relevance of clustering ICP time-series data and investigate their relationship with CT findings.*

Steps required:

First objective → *apply and compare a set of both standard feature based and waveform-based clustering approaches, capable of grouping patients according to the shape and evolution of their 24-hour ICP profile, without relying on any predefined clinical thresholds or outcome labels.*

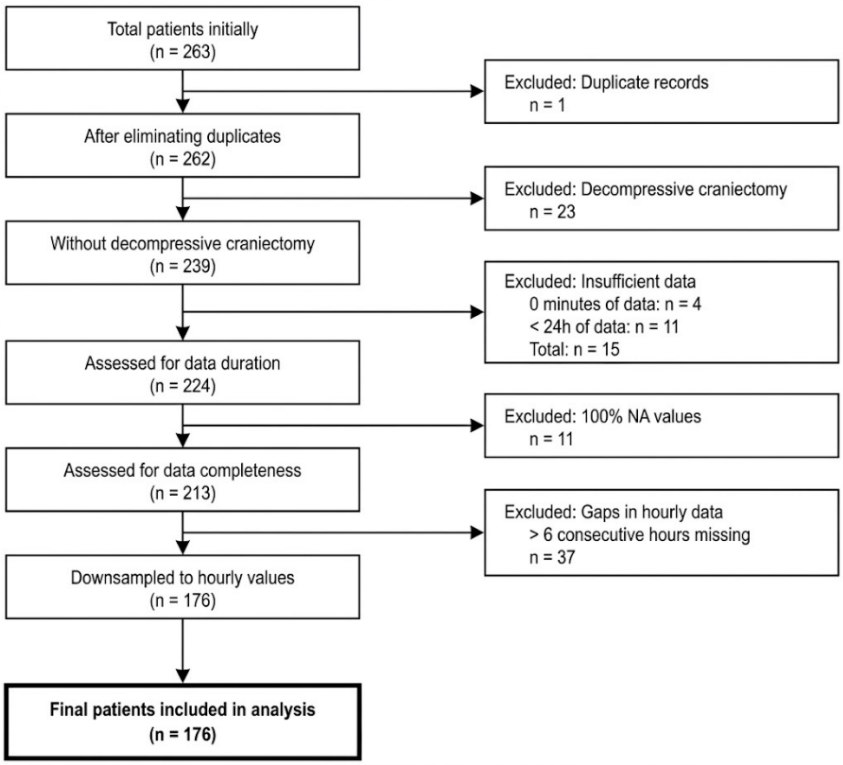
If the results show patterns between the ICP profiles after the clustering, then:

Second objective → *evaluate clinically the resulting clusters by examining their associations with CT scan findings. Specifically, the Marshall CT classification recorded at admission is used as the primary external variable against which the ICP clusters identified through data driven methods are evaluated, with the goal of determining whether distinct pressure evolution patterns are systematically associated with specific patterns of structural brain pathology, moving toward a more physiologically grounded understanding of the relationship between the damage visible on imaging and the pressure behavior observed in the patient.*

DATA ENGINEERING & SIGNAL PROCESSING PIPELINE

The project used the European multicenter **BrainIT database**, which tracks high-frequency neuro-intensive care monitoring data across European centers. The cohort initially contained 263 patient records pairing continuous minute-by-minute physiological time-series with static clinical and neuroimaging variables.

Signal Processing Overview



Clinical Preprocessing & Plausibility Filtering

- **Preserving Closed-Cranial Physics:** Patients who underwent decompressive craniectomy were excluded because removing cranial bone transforms the cranium from a closed to an open compartment, fundamentally altering compliance and invalidating pressure-volume comparisons.
- **Window Standardization:** Patient signals were anchored to the first 24 hours post-ICU admission. Incomplete windows were removed, and longer recordings were truncated to maintain an identical observation baseline.
- **Physiological Plausibility Mask:** Measurements outside clinical limits (< 0 mmHg or > 100 mmHg) were flagged as recording errors and converted to missing values.

Robust Non-Stationary Denoising & Reconstruction

Clinical ICU environments introduce high-frequency noise from patient movements, coughing, arterial line flushes, and sensor handling. Therefore, the step of artifact removal (artifact detection + artifact imputation) is necessary to avoid keeping false spikes or removing genuine physiological ones.

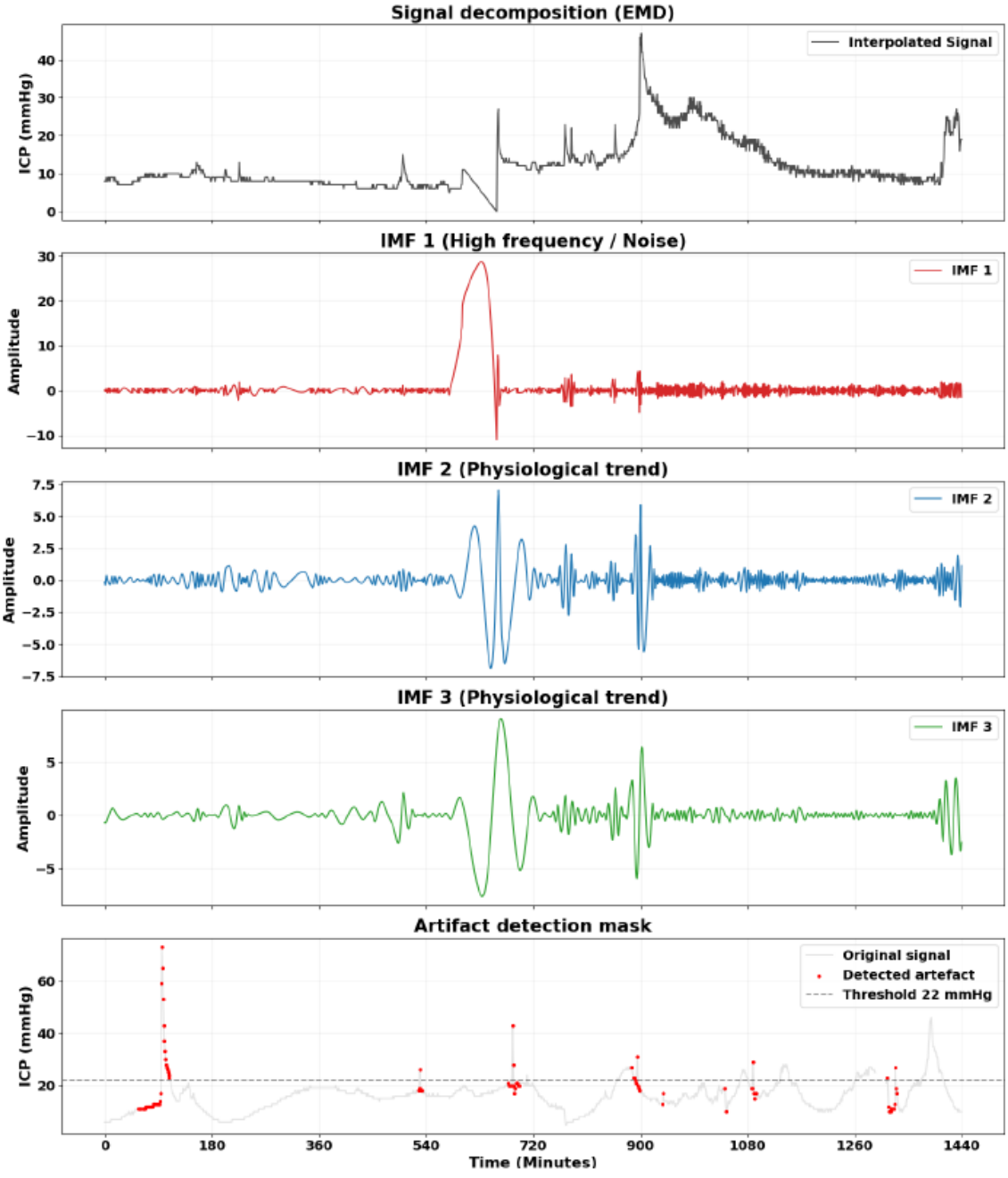
1. **Empirical Mode Decomposition (EMD):** Because ICP dynamics are nonlinear and non-stationary, EMD was used to decompose each raw signal into Intrinsic Mode Functions (IMFs). High-frequency transient artifacts concentrate within the first component (IMF1), whereas slower, clinically vital trends remain preserved in subsequent components (IMF2, IMF3...).

2. **Robust Statistical Thresholding:** Standard deviations are biased by extreme outliers. Instead, a robust threshold was established on IMF1 using Median Absolute Deviation (MAD), setting $\sigma_{\text{MAD}} = 1.4826 \times \text{MAD}$. Data points exceeding $3\sigma_{\text{MAD}}$ were masked as corrupted.

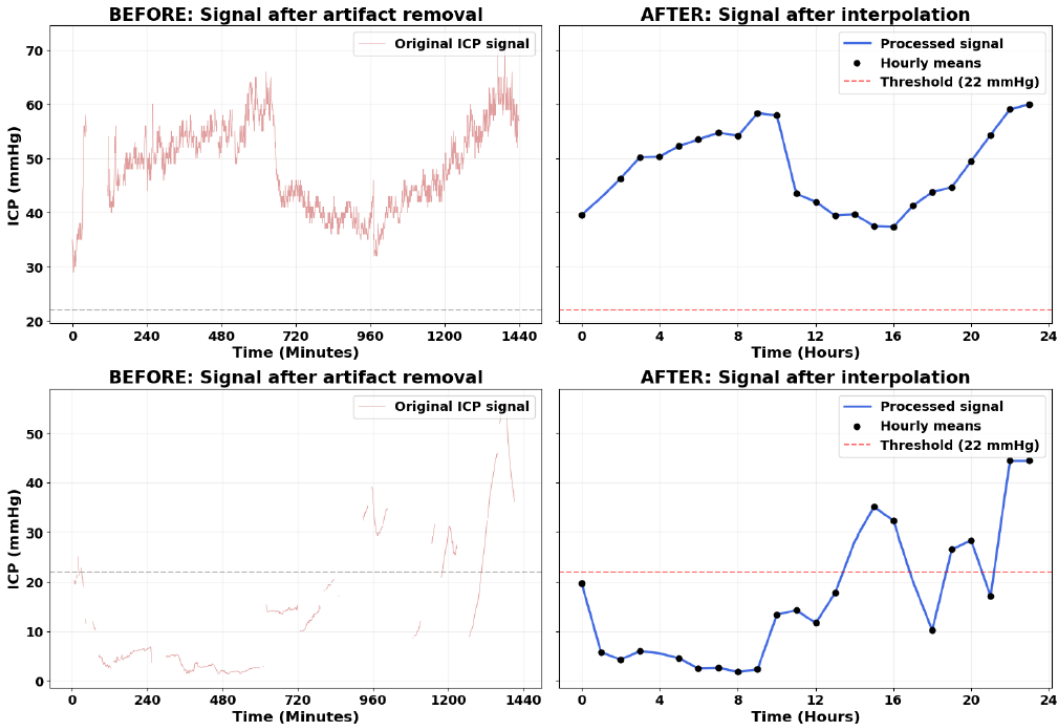
3. **Hourly Aggregation & Gap Control:** Signals were downsampled to hourly resolution, requiring at least 20 validated artifact-free minutes to compute a reliable hourly mean. Patients with consecutive missing intervals exceeding 6 hours were excluded to prevent interpolation bias, yielding a validated cohort of 176 patients (retention $> 80\%$).

4. **PCHIP Imputation:** Missing intervals across the standardized 24-point profiles were reconstructed using Piecewise Cubic Hermite Interpolating Polynomials (PCHIP). Unlike cubic splines that overshoot and introduce spurious peaks into steep physiological slopes, PCHIP guarantees local monotonicity and respects biological boundaries.

Artifact Detection Overview



Signal before vs. after artifact imputation for two patients



FEATURE EXTRACTION & SELECTION

To enable feature-based clustering algorithms to process the temporal recordings, feature extraction was done to capture the most clinically and statistically relevant characteristics of the signal. This resulted in **165 dynamic features** grouped across several domains.

Integration of Static Variables & Feature Selection:

- Static admission features (Age, Sex, Pupil Reactivity, etc.) were integrated. Pupil reactivity was transformed via One-Hot Encoding to avoid imposing artificial ordinal linearity between categorical neurological states.
- Features were standardized using Z-score normalization.
- A pairwise Pearson correlation filter removed collinear variables with $|r| > 0.90$
- A pairwise Mutual Information (MI) filter removed features with average redundant scores > 0.10, consolidating the matrix into **76 independent, informative features**.
- Principal Component Analysis (PCA) revealed that **26 components were required to reach 80% explained variance**, proving that the underlying physiological patterns were distributed without a single dominant factor.

UNSUPERVISED CLUSTERING

The following clustering algorithms were tested across the patient cohort:

- Standard approaches
  - K-means
  - DBSCAN
  - Spectral
  - GMM
- Waveform-based approaches
  - SOM
  - DTW

Candidate Selection for Clinical Evaluation

Each clustering methodology was evaluated to assess whether it could produce meaningful, stable patient groupings. Certain approaches were discarded because they proved unviable for this dataset: DBSCAN failed due to the continuous nature of the ICP distribution, resulting either in a single over-generalized cluster or classifying peripheral patients as noise, while Spectral clustering showed substantial overlap in the reduced space and weak practical separation despite parameter tuning. For the viable algorithms, candidate solutions were carried forward into the next stage to determine whether the clusters captured clinically relevant structure rather than arbitrary partitions.

To evaluate whether these candidate solutions were associated with clinically relevant variables, their relationship with the Marshall CT classification was quantified using **Cramér's V** at two critical milestones: admission and the worst CT time point. According to these results, the top-performing algorithms across the cohort were **SOM**, **DTW**, and **K-Means**. The detailed association analysis focused specifically on the configurations demonstrating the strongest and most consistent Cramér's V values across both time points: SOM ($k = 8$), DTW ($k = 8$), and K-Means ($k = 6$).

CT ASSOCIATION ANALYSIS & PHENOTYPE CHARACTERIZATION

To interpret the clinical relevance of the selected cluster configurations, two visualization tools were used:

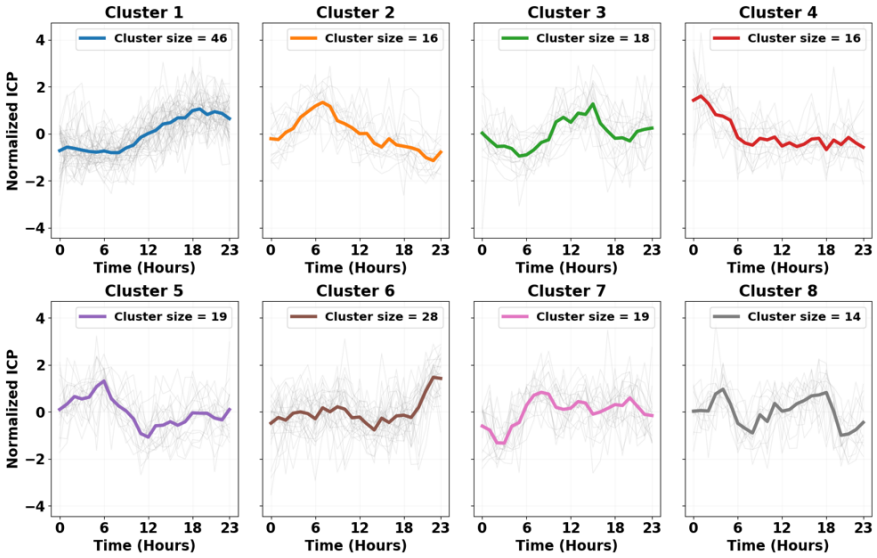
- **Marshall CT Heatmaps** → Numerical Cramér's V values confirm the statistical strength of an association, but they do not reveal *which* specific injuries drive that relationship. Heatmaps cross-referenced cluster membership directly against the six Marshall CT categories at both admission and the worst CT time point. Comparing these two time points made it possible to track anatomical evolution over time, identifying for example, whether an ICP trajectory was linked to expanding focal mass lesions or a shift toward diffuse swelling.
- **Deviation from Population Mean Graphs** → A complete clinical phenotype requires looking beyond imaging alone. Deviation graphs benchmarked each cluster against the cohort baseline across static neurological indicators (age, sex, admission GCS motor score, pupil reactivity) and 6-month clinical outcomes (ICU mortality and GOSe functional recovery tiers). By standardizing values into Z-scores (red for higher than average, blue for lower) and applying **matrix seriation** (reordering rows and columns via hierarchical clustering so related features cluster together visually), these graphs transformed abstract mathematical groups into clear, interpretable clinical phenotypes.

RESULTS

While alternative waveform-based methods like DTW also demonstrated promising results, the analysis below focuses specifically on the Self-Organizing Map (SOM, k=8) solution, which produced the clearest trajectory patterns and the strongest, most consistent clinical associations across both time points.

SOM cluster results for k=8

- Cluster 2 **(Early Rise & Stabilization)** → Peaks within the first 8 hours then steadily declines; captures acute focal lesions successfully stabilized early.
- Cluster 3 **(Late Escalation)** → Remains stable for ~5 hours before entering a delayed crisis. Older cohort with the highest ICU mortality and severe disability.
- Cluster 5 **(Acute Early Spike)** → Peaks around hour 5, then resolves quickly. Younger cohort, with preserved pupil reactivity and favorable survival.
- Cluster 6 **(Delayed Rise)** → Stable baseline followed by a sharp terminal rise. Reflects moderate diffuse injuries progressing toward secondary mass lesions.
- Cluster 8 **(Prolonged Instability)** → Continuous oscillating peaks between hours 3 and 18. Captures silent physiological deterioration and rising intracranial stiffness.



To bring all this together, the table below summarizes the SOM phenotypes and highlights where static imaging and dynamic trajectories diverge. The central finding is that structural severity on CT does not always correlate with clinical outcome:

- Cluster 2 **(Focal injury with high survival)** → Presents high-grade mass lesions ($45 \rightarrow 60\%$) and severely impaired pupils, yet achieved **0% ICU mortality** following early pressure stabilization.
- Cluster 3 **(Diffuse injury with poor outcome)** → Shows moderate diffuse damage with relatively preserved pupils, yet suffers the **worst clinical outcomes** (highest mortality and vegetative rate).
- Cluster 8 **(Silent deterioration)** → Presents the **healthiest admission exam** (best motor scores and 0% non-reacting pupils), yet its volatile trajectory uncovers a silent progression into severe brain swelling (Diffuse 3).

These dissociations demonstrate that dynamic ICP waveforms capture critical physiological risks that static CT scans and initial exams miss entirely.

Cluster	Trajectory	CT / structure	Outcome	Phenotype
2	Early rise, then decline	Mass lesions 45 → 60%	ICU mortality 0%	Severe focal, high survival
3	Late escalation	Moderate diffuse (diffuse 2)	Worst → high mortality & disability	Diffuse, poor outcome
5	Mid spike, then settles	Mildest diffuse, low mass lesion	Favorable survival	Mild diffuse, good survival
6	Stable, then late rises	Diffuse → mass lesion (progresses)	Survives but lasting disability	Delayed focal progression
8	Prolonged instability	Diffuse 2 → diffuse 3 (worsens)	Death rate above average	Silent deterioration

CONCLUSIONS

Working with the multicenter BrainIT cohort, this project developed a pipeline to process continuous 24-hour ICP signals and evaluate whether unsupervised clustering methods (K-means, SOM, DTW, etc.) could uncover patient phenotypes that static thresholds miss. The pipeline combined EMD and MAD filtering to strip bedside noise while using PCHIP interpolation to reconstruct missing intervals without distorting signal shape. Comparing the resulting clusters against Marshall CT scans using Cramér's V highlighted key discrepancies between anatomical damage on imaging and real-time physiological deterioration. Presenting these patterns through heatmaps and deviation graphs made the complex groupings straightforward for clinical teams to interpret, demonstrating the value of tracking full trajectory shapes rather than relying solely on isolated pressure alarms. Ultimately, this approach offers a practical foundation for shifting neuro-intensive care from reactive alarm monitoring toward proactive, trajectory-guided clinical decision support.